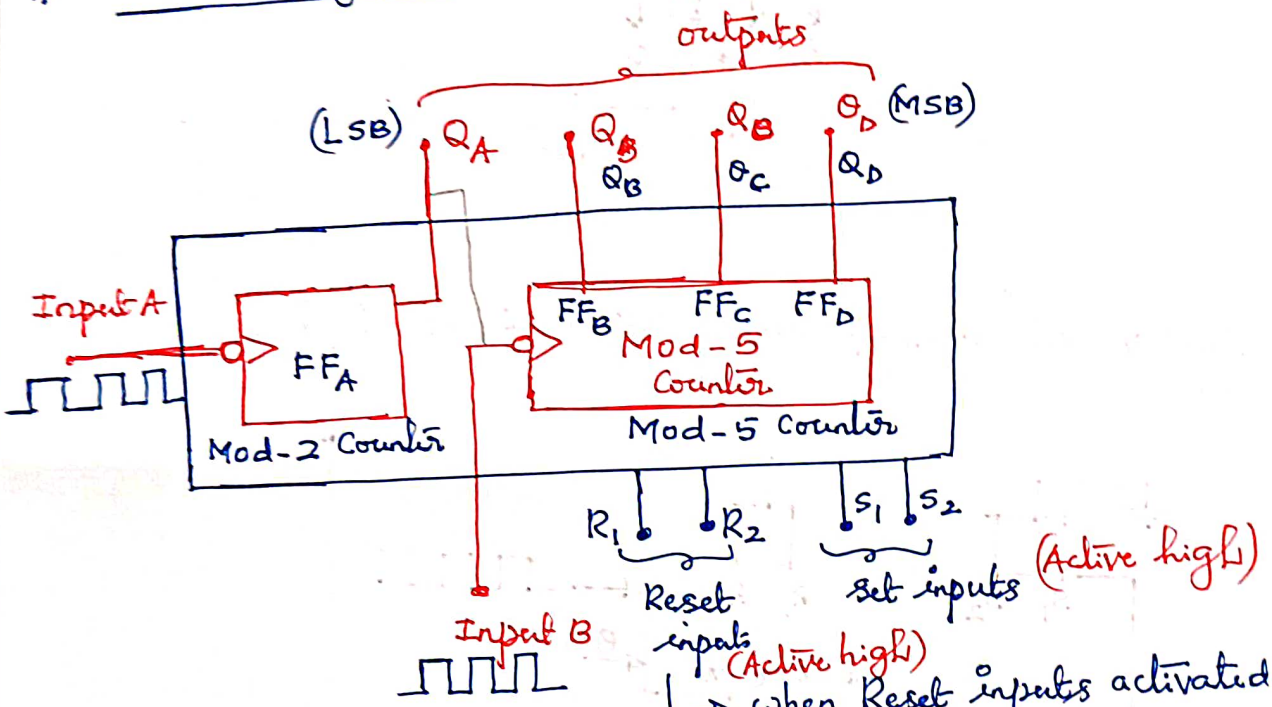
IC 7490 Asynchronous Counter

↳ IC 7490 is a ripple decade counter, also known as MOD-10 counter.

↳ It has MOD-2 counter and MOD-5 counter

* Block diagram of IC 7490:

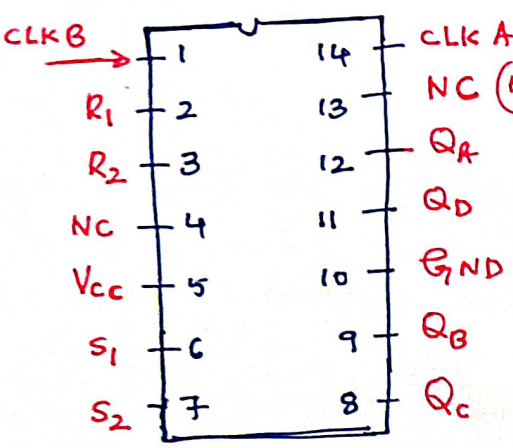


↳ when Reset inputs activated, the o/p's of FF's is cleared i.e 0.

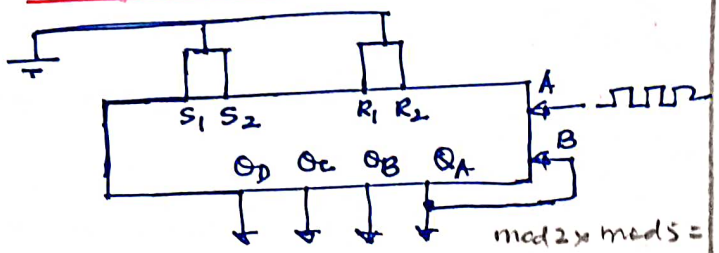
↳ When set inputs activated, then o/p is set to

Q_D (MSB)	Q_C	Q_B	Q_A (LSB)
1	0	0	1

Pin diagram of 7490

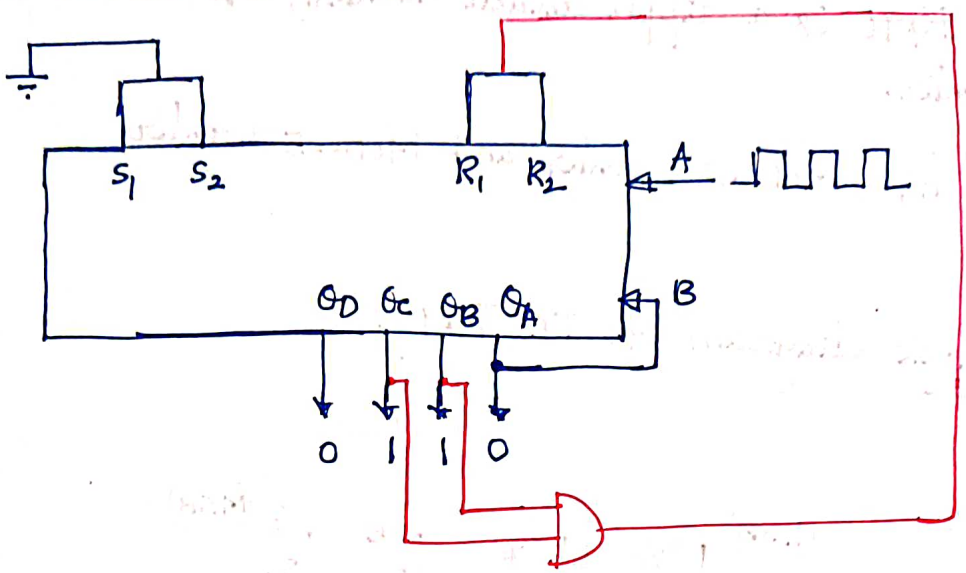


Design of MOD-10 counter



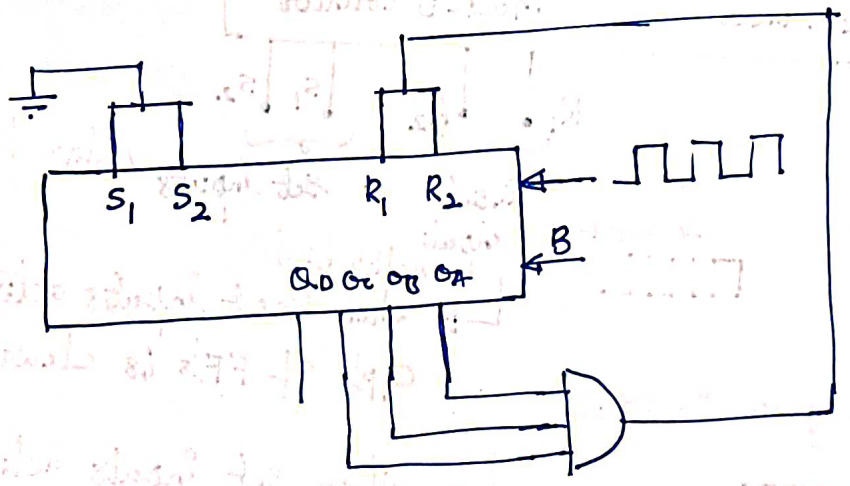
mod 2 × mod 5 = 10 mod

* Design of MOD-6 Counter



0000
0001
0010
0011
0100
0101
0110 → Reset

* Design of MOD-7 Counter using IC 7490:

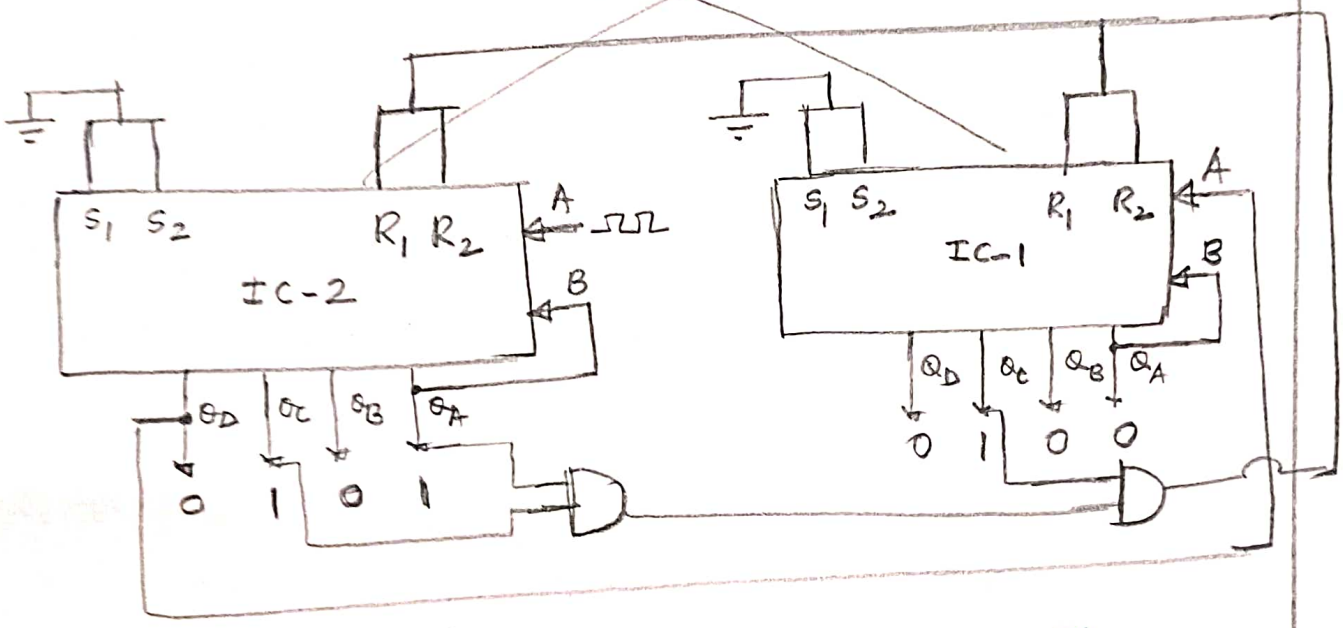


* Design of Mod-45 counter using 7490:

0000
0001
0010
0011
0100
0101

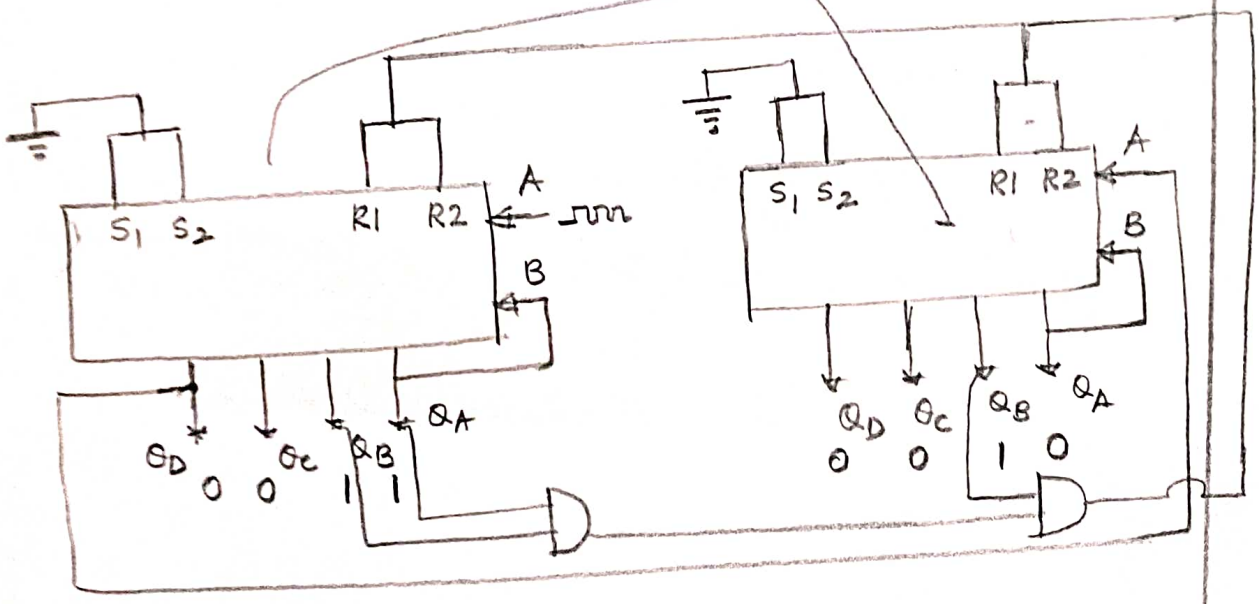
Example1: 45 ⟶

Ten's place	Unit place
4	5
0100	0101
$Q_D\ Q_C\ Q_B\ Q_A$	$Q_D\ Q_C\ Q_B\ Q_A$
IC-1	IC-2



Example: 23 ⟶

2	3
0010	0011
$Q_D\ Q_C\ Q_B\ Q_A$	$Q_D\ Q_C\ Q_B\ Q_A$

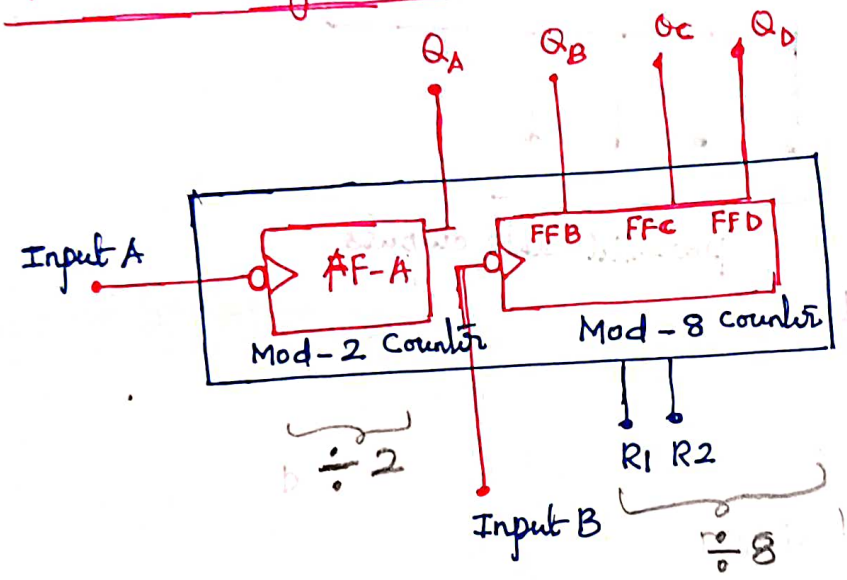


* IC 7493 : Counter Connection [4-bit binary counter]

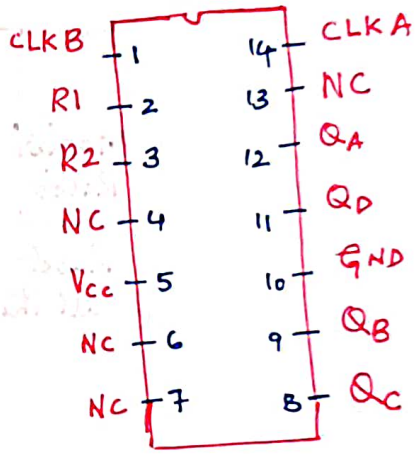
↳ The 7493 IC is an up-counter that is capable of operating as a multi-modulus counter

↳ It is constructed of two Negative-edge triggered counters in their natural state are

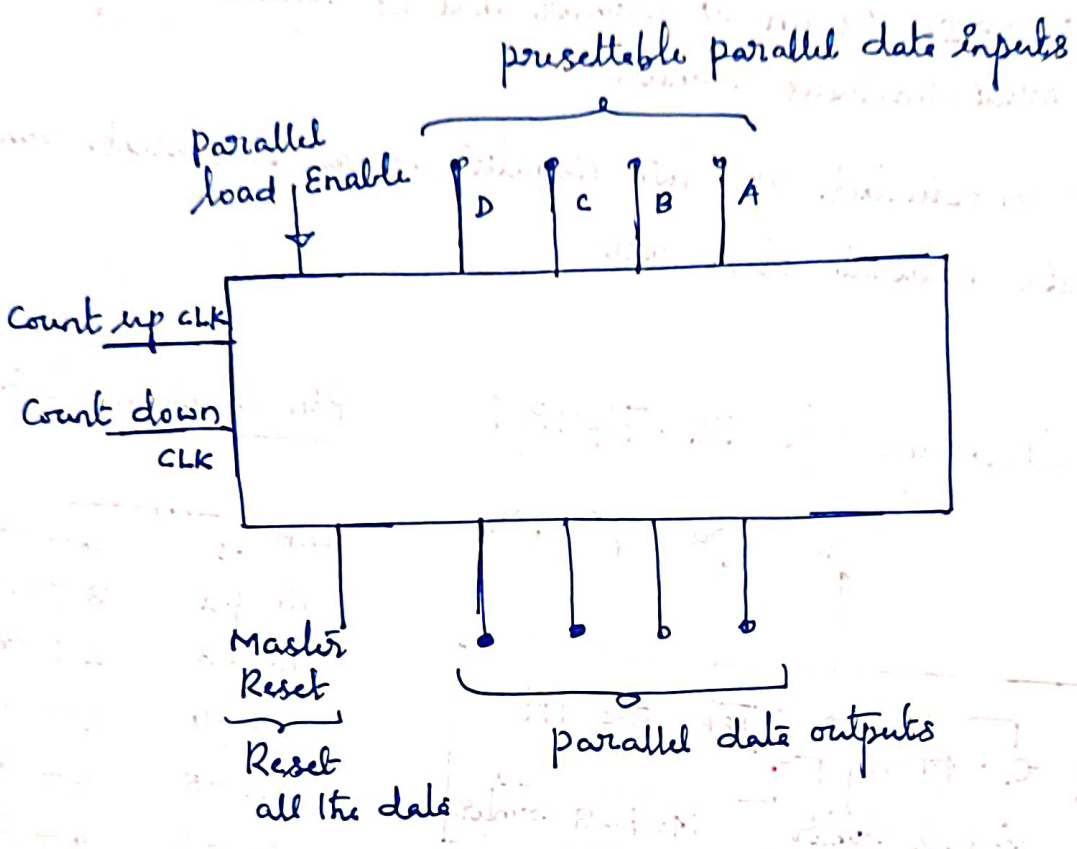
* Block diagram of IC-7493 :



Pin diagram of 7493



R_1R_2 → when logic highs are applied to both reset leads simultaneously, the count resets to 0000.

→ under normal counts, both leads should be connected to logic low.

IC - 74192 : 4 - Bit Binary decade up/down Counter